Nepal TST 2025 Solutions

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§1 Day 1 Problems

Problem 1.1. Consider a triangle $\triangle ABC$ and some point X on BC. The perpendicular from X to AB intersects the circumcircle of $\triangle AXC$ at P and the perpendicular from X to AC intersects the circumcircle of $\triangle AXB$ at Q. Show that the line PQ does not depend on the choice of X.

Problem 1.2. Find all integers n such that if

$$1 = d_1 < d_2 < \cdots < d_{k-1} < d_k = n$$

are the divisors of n, then the sequence

$$d_2 - d_1,\ d_3 - d_2,\ \ldots,\ d_k - d_{k-1}$$

forms a permutation of an arithmetic progression.

Problem 1.3. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(f(x)) + xf(xy) = x + f(y)$$

for all positive real numbers x and y.

§2 Day 2 Problems

Problem 2.1. Let the sequence $\{a_n\}_{n\geq 1}$ be defined by

$$a_1 = 1, \quad a_{n+1} = a_n + \frac{1}{\sqrt[2024]{a_n}} \quad \text{for } n \geq 1,\, n \in \mathbb{N}$$

Prove that

$$a_n^{2025} > n^{2024}$$

for all positive integers $n \geq 2$.

Problem 2.2. Kritesh manages traffic on a 45×45 grid consisting of 2025 unit squares. Within each unit square is a car, facing either up, down, left, or right. If the square in front of a car in the direction it is facing is empty, it can choose to move forward. Each car wishes to exit the 45×45 grid.

Kritesh realizes that it may not always be possible for all the cars to leave the grid. Therefore, before the process begins, he will remove k cars from the 45×45 grid in such a way that it becomes possible for all the remaining cars to eventually exit the grid.

What is the minimum value of k that guarantees that Kritesh's job is possible?

Problem 2.3. Consider an acute triangle ΔABC. Let D and E be the feet of the altitudes from A to BC and from B to AC respectively.

Define D_1 and D_2 as the reflections of D across lines AB and AC, respectively. Let Γ be the circumcircle of ΔAD_1D_2. Denote by P the second intersection of line D_1B with Γ, and by Q the intersection of ray EB with Γ.

If O is the circumcenter of ΔABC, prove that O, D, and Q are collinear if and only if quadrilateral $BCQP$ can be inscribed within a circle.

§3 Solutions to Day 1

Problem 3.1. Consider a triangle $\triangle ABC$ and some point X on BC. The perpendicular from X to AB intersects the circumcircle of $\triangle AXC$ at P and the perpendicular from X to AC intersects the circumcircle of $\triangle AXB$ at Q. Show that the line PQ does not depend on the choice of X.

Proof. Denote by O the circumcenter of $\triangle ABC$. We show that PQ always lies on the line AO, equivalent to showing $A - O - P$ and $A - O - Q$. Let D be the point on AC such that $XD \perp AC$.

Claim 3.2 — We have $A - O - Q$.

Proof.

$$\measuredangle BAO = \frac{180^\circ - \measuredangle AOB}{2} = 90^\circ - \measuredangle ACB = \measuredangle CXD = \measuredangle BAQ$$

□

Claim 3.3 — We have $A - O - P$.

Proof.

$$\measuredangle CAP = \measuredangle CXP = 90^\circ + \measuredangle CBA = \measuredangle CAO$$

□

Therefore, P and Q lie on AO, we're done. □

Problem 3.4. Find all integers n such that if

$$1 = d_1 < d_2 < \cdots < d_{k-1} < d_k = n$$

are the divisors of n, then the sequence

$$d_2 - d_1,\ d_3 - d_2,\ \ldots,\ d_k - d_{k-1}$$

forms a permutation of an arithmetic progression.

Proof. □

Problem 3.5. Find all functions $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that

$$f(f(x)) + xf(xy) = x + f(y)$$

for all positive real numbers x and y.

Proof. Let $P(x, y)$ be the given assertion. We claim that the only solutions for f are $f(x) = 1$ and $f(x) = \frac{c}{x}$ for some constant c. These work upon checking.

When f is constant, we get $f(x) = 1$, so from henceforth assume that f is not constant.

$$P(1, x) \implies f(f(1)) + f(x) = 1 + f(x) \implies f(f(1)) = 1$$

$$P(x, f(x)) \implies f(f(x)) + xf(xf(x)) = x + f(f(x)) \implies f(xf(x)) = 1$$

Therefore, if f was injective, we would have $xf(x) = f(1)$ and we would be done. So, we prove the following claim. □

§4 Solutions to Day 2

Problem 4.1. Let the sequence $\{a_n\}_{n\geq 1}$ be defined by

$$a_1 = 1, \quad a_{n+1} = a_n + \frac{1}{\sqrt[2024]{a_n}} \quad \text{for } n \geq 1,\, n \in \mathbb{N}$$

Prove that

$$a_n^{2025} > n^{2024}$$

for all positive integers $n \geq 2$.

> **Remark 4.2.** In my opinion, this is the absolute easiest problem of the six. This problem is easily solved using induction, as we will see below. I've also heard/seen about solutions using calculus, but I think most of them are inductive too, so it seems induction is the way to go for this problem.

Proof. Note that for all k, $a_{k+1} > a_k$. We proceed via induction.

The base case $n = 2$ is obvious as $a_2^{2025} = 2^{2025} > 2^{2024}$ holds. Assume now, for some k that $a_k^{2025} > n^{2024}$. It suffices to show that $a_{k+1}^{2025} > n^{2024}$ or equivalently, $a_{k+1}^{\frac{2025}{2024}} > k+1$. This is easy, however, as:

$$\begin{aligned} a_{k+1}^{\frac{2025}{2024}} &= a_{k+1} \cdot a_{k+1}^{\frac{1}{2024}} \\ &> a_{k+1} \cdot a_k^{\frac{1}{2024}} \\ &= \left(a_k + \frac{1}{a_k^{\frac{1}{2024}}} \right) a_k^{\frac{1}{2024}} \\ &= a_k^{\frac{2025}{2024}} + 1 \\ &> k+1 \end{aligned}$$

And we're done. □

Problem 4.3. Kritesh manages traffic on a 45×45 grid consisting of 2025 unit squares. Within each unit square is a car, facing either up, down, left, or right. If the square in front of a car in the direction it is facing is empty, it can choose to move forward. Each car wishes to exit the 45×45 grid.

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What is the minimum value of k that guarantees that Kritesh's job is possible?

Proof. □

Problem 4.4. Consider an acute triangle ΔABC. Let D and E be the feet of the altitudes from A to BC and from B to AC respectively.

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Proof. □